



**Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 6 • STANDARD 6.AT.7

# Simplify Algebraic Expressions

Lesson 6-15 • Teacher Copy

**Name:** \_\_\_\_\_

**Date:** \_\_\_\_\_

**Class:** \_\_\_\_\_

## My Notes

Level 2 Standard



### TODAY'S OBJECTIVES

**Content Objective:** I can simplify algebraic expressions by combining like terms.

**Language Objective:** I can explain my steps using the words like terms, combine, simplify, and coefficient.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



**Fill in each blank as we go. Use the Word Bank to help you.**



### WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Like Terms

Combine

Simplify

Coefficient

Term

Distributive property



Tap any word to see its meaning and a picture.

1

Terms with the same variable part, like  $3y$  and  $8y$ , are

.

2

To add or subtract like terms into a single term is to

them.

3

Rewriting an expression in a shorter form that is still equal to it is to

it.

4

The number multiplied by a variable is the  .

5

A number or variable separated by  $+$  or  $-$  signs is a

.

- 6 Multiplying each part of a sum by a number uses the  property.

## Write About the Math

**Simplify the total order expression. What does each part of the simplified expression tell you?**

*Simplifica la expresión del pedido total. ¿Qué te dice cada parte de la expresión simplificada?*

Say your answer to a partner first. Then write it, using at least two of these words:

☐

**Like Terms**

*Términos semejantes*

☐

**Combine**

*Combinar*

☐

**Simplify**

*Simplificar*

☐

**Coefficient**

*Coeficiente*

☐

**Term**

*Término*

**Model:** *You can only combine items of the same type, just like like terms.* — Students explain microphones combine with microphones ( $4 + 2 + 6 = 12$ ) but not with stands, connecting this to combining like terms in algebra.

### Start

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- **The total order simplifies to** \_\_\_\_ **m +** \_\_\_\_ **. The** \_\_\_\_ **tells us** \_\_\_\_ **and the** \_\_\_\_ **tells us** \_\_\_\_.
- **My answer is** \_\_\_\_ **because** \_\_\_\_.

*Mi respuesta es* \_\_\_\_ *porque* \_\_\_\_.

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### Build

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- **My claim is** \_\_\_\_.
- **My evidence is** \_\_\_\_ **, so** \_\_\_\_.

*Mi afirmación es* \_\_\_\_.

*Mi evidencia es* \_\_\_\_ *, entonces* \_\_\_\_.

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### Explain

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

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☐ I answered the question.

☐ I used math evidence: numbers, an equation, or a model.

☐ I used at least two math words.

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## Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

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## Answer Key & Teacher Guide

1. **Notes 1:** Like Terms
2. **Notes 2:** Combine
3. **Notes 3:** Simplify
4. **Notes 4:** Coefficient
5. **Notes 5:** Term
6. **Notes 6:** Distributive property
7. **Watch for this mistake:** *A common mistake in Simplify Algebraic Expressions is sweeping a constant into the variable terms — for example, simplifying  $6x + 4 + 2x$  as  $12x$  by adding all three numbers ( $6 + 4 + 2 = 12$ ) instead of recognizing that 4 has no variable and cannot combine with  $6x$  and  $2x$ . The correct simplified expression is  $8x + 4$  (combine  $6x + 2x = 8x$ , and let the 4 stay on its own). Before you submit, check that every term you combined has the exact same variable part.*
8. **Try It 1:** A.  $4y$  — Subtract the coefficients of the like terms:  $7 - 3 = 4$ . Keep the variable  $y$ , giving  $4y$ .
9. **Try It 2:** A.  $9m + 4 - 6m + 3m = 9m$ . The constant 4 has no like term, so it stays. Result:  $9m + 4$ .